COGNIZANT SUPERSET ID : 6424916

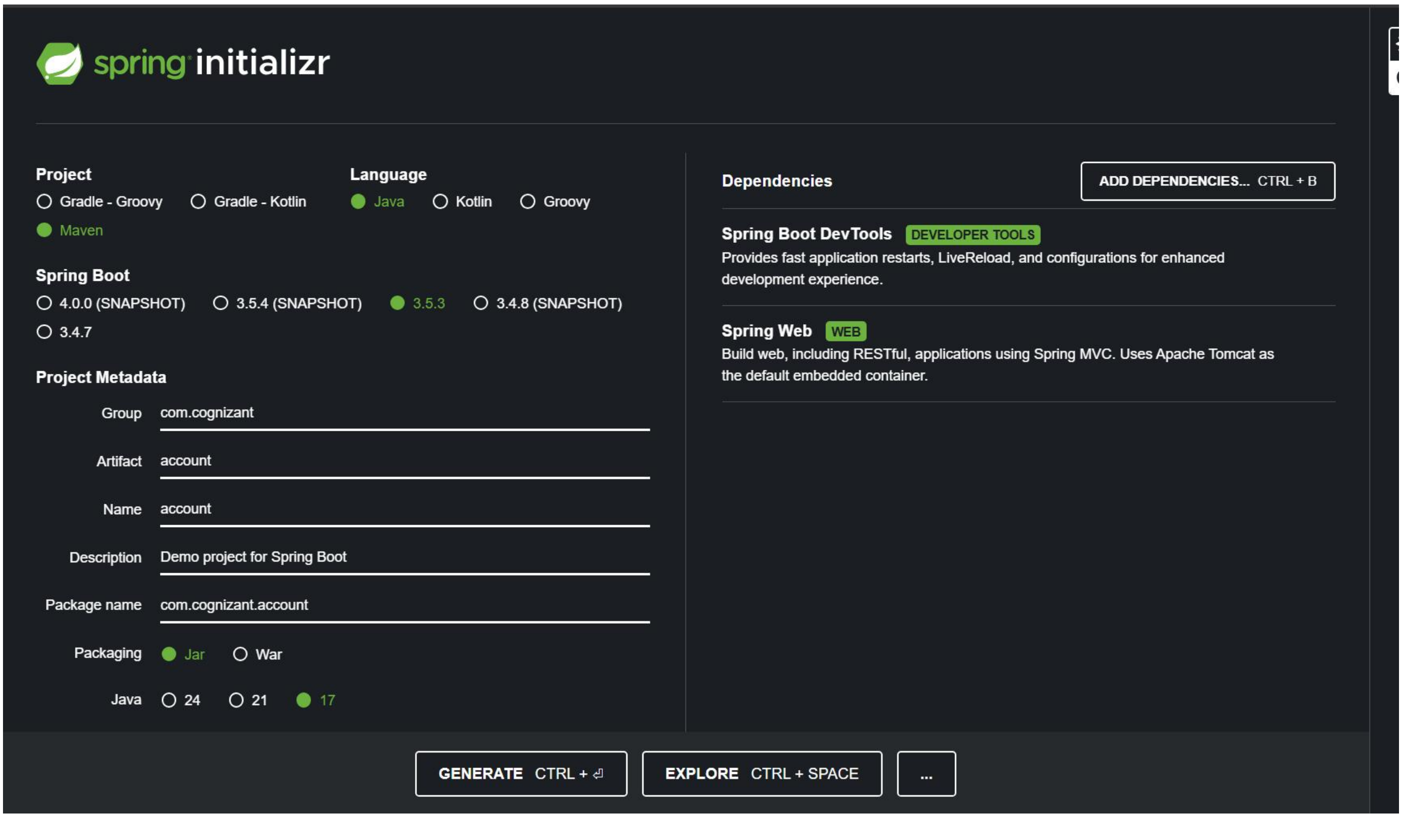
2. Microservices with API gateway

Creating Microservices for account and loan

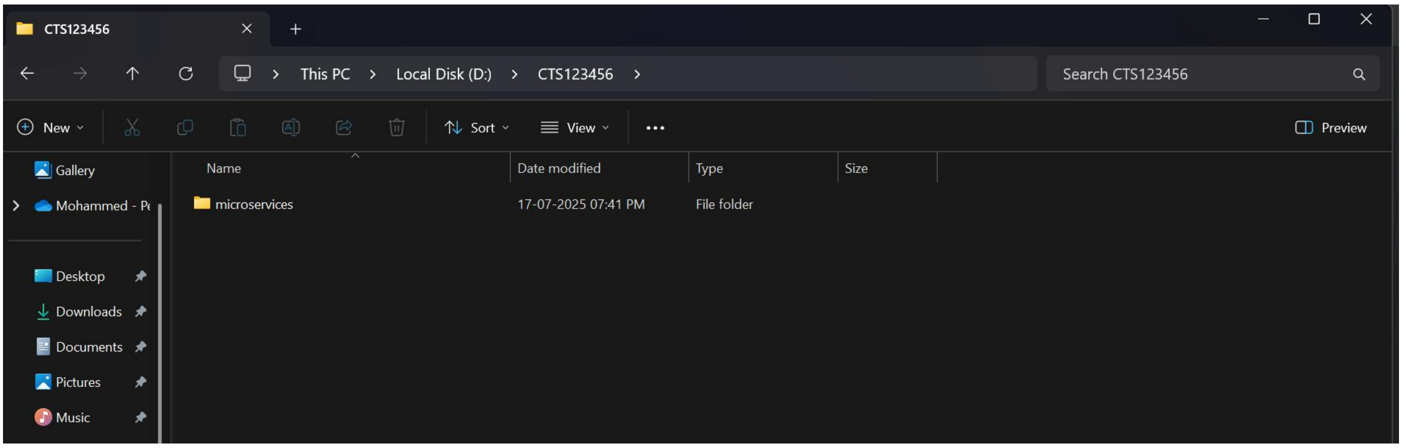
Account Microservice

- Create folder with employee id in D: drive
- Create folder named 'microservices' in the new folder created in previous step. This folder will contain all the sample projects that we will create for learning microservices.
- Open https://start.spring.io/ in browser
- Enter form field values as specified below:
  - **Group:** com.cognizant
  - **Artifact:** account
- Select the following modules
  - Developer Tools > Spring Boot DevTools
  - Web > Spring Web
- Click generate and download the zip file
- Extract 'account' folder from the zip and place this folder in the 'microservices' folder created earlier
- Open command prompt in account folder and build using mvn clean package command
- Import this project in Eclipse and implement a controller method for getting account details based on account number. Refer specification below:
  - Method: GET
  - Endpoint: /accounts/{number}
  - Sample Response. Just a dummy response without any backend connectivity.

```
{ number: "00987987973432", type: "savings", balance: 234343 }
```

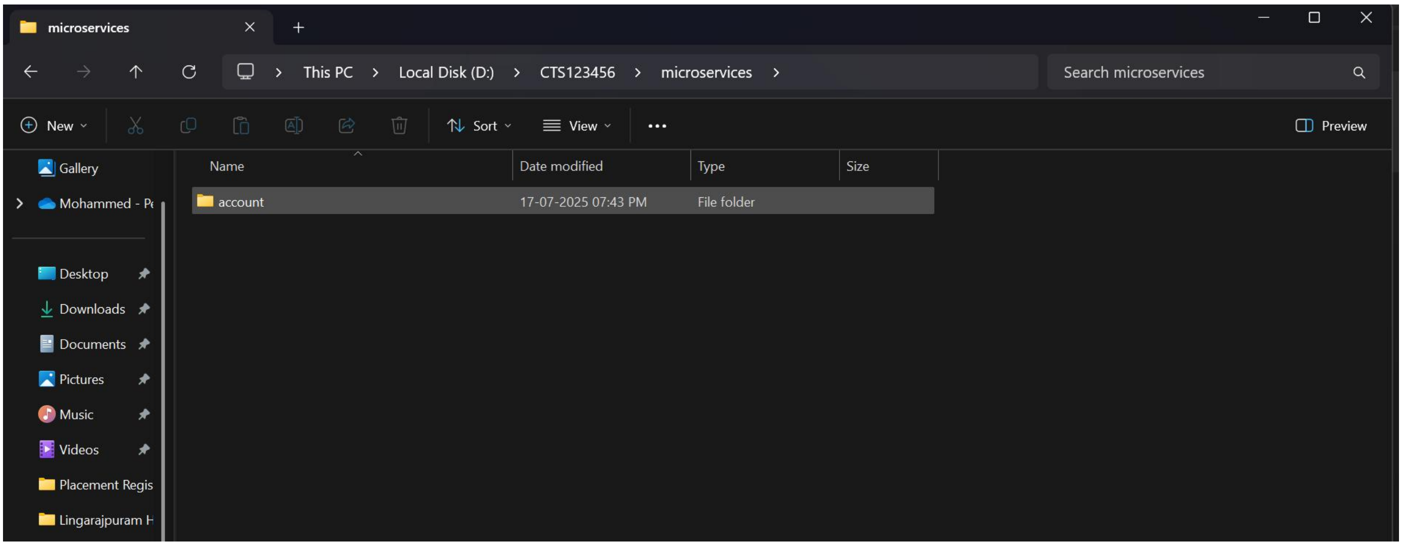


Create a folder in D drive

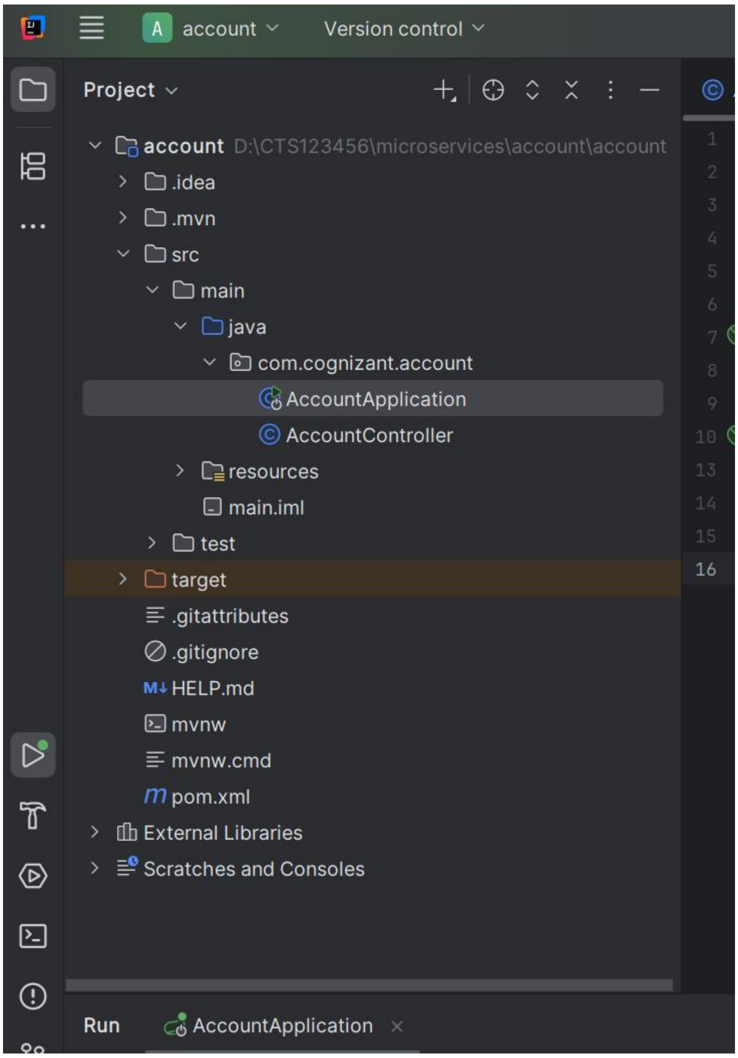


Create a folder named microservices inside the created folder

Then, extract the spring initializr file and paste it inside the "microservices" folder



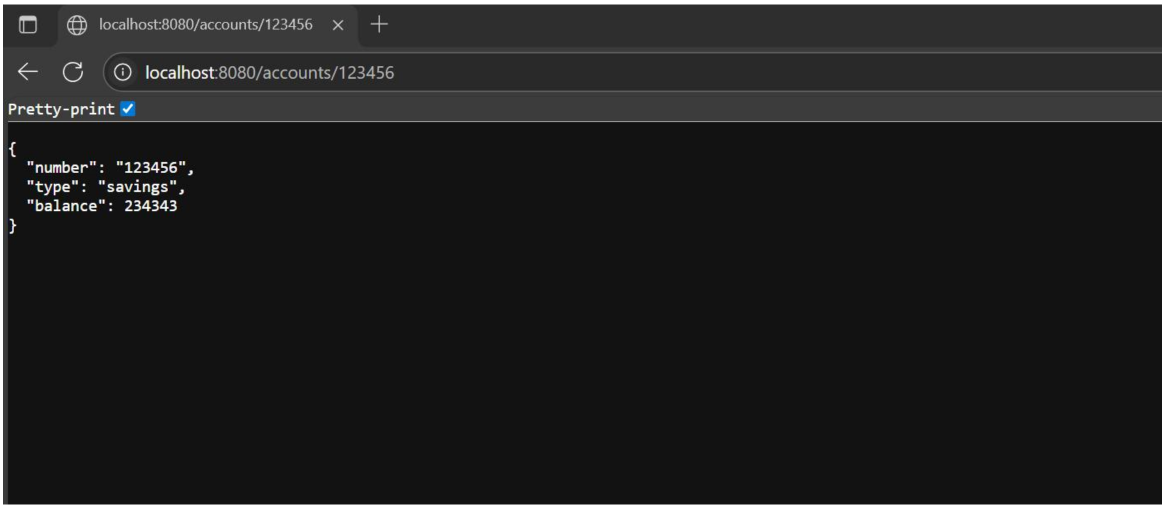
Folder Structure:



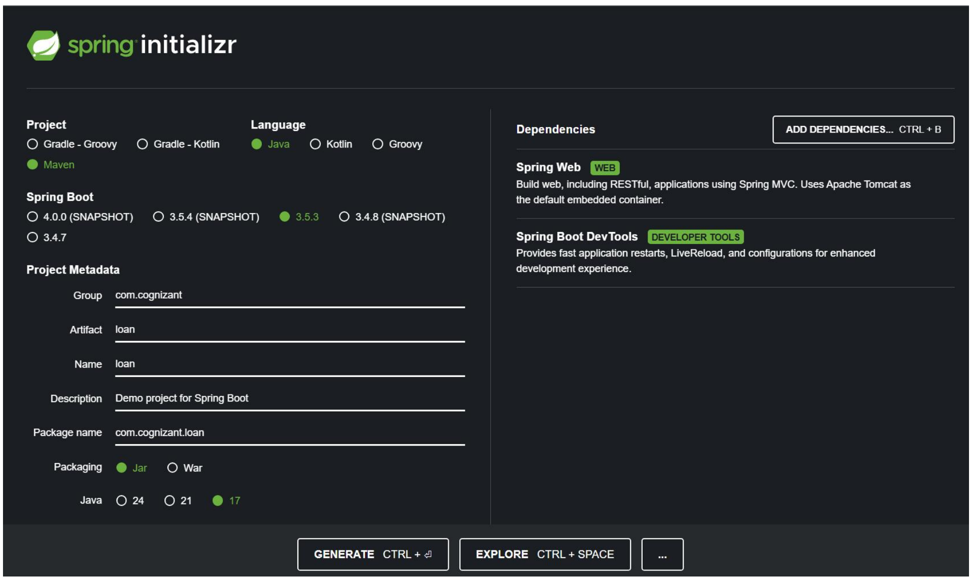
AccountController.java

```java
package com.cognizant.account;

import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/accounts")
public class AccountController {

    @GetMapping("/{number}")
    public Account getAccount(@PathVariable String number) {
        return new Account(number, "savings", 234343);
    }

    record Account(String number, String type, double balance) {}
}
```

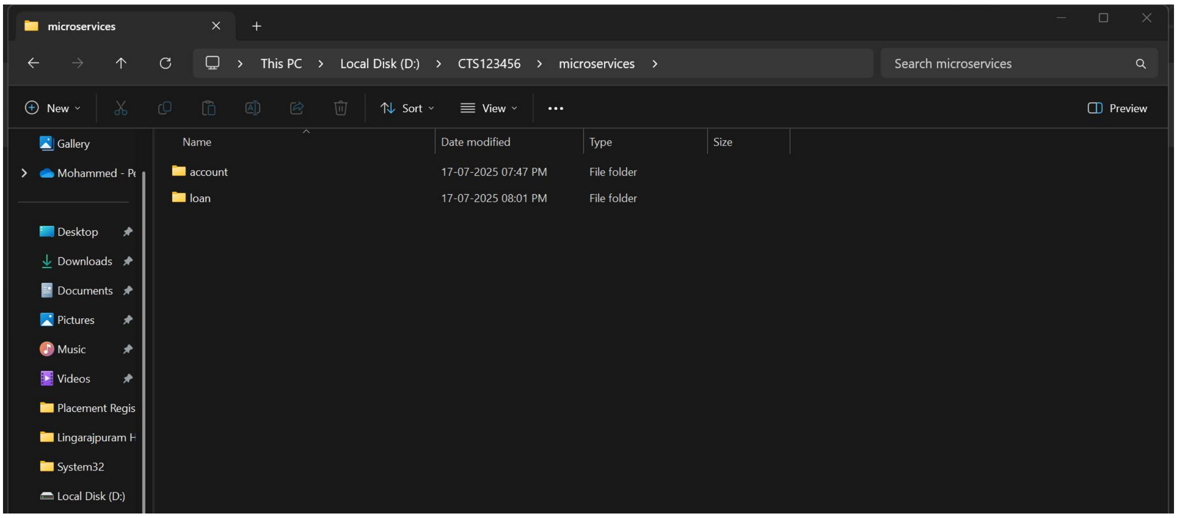
Then, run AccountApplication.java and go to localhost:8080/accounts/123456



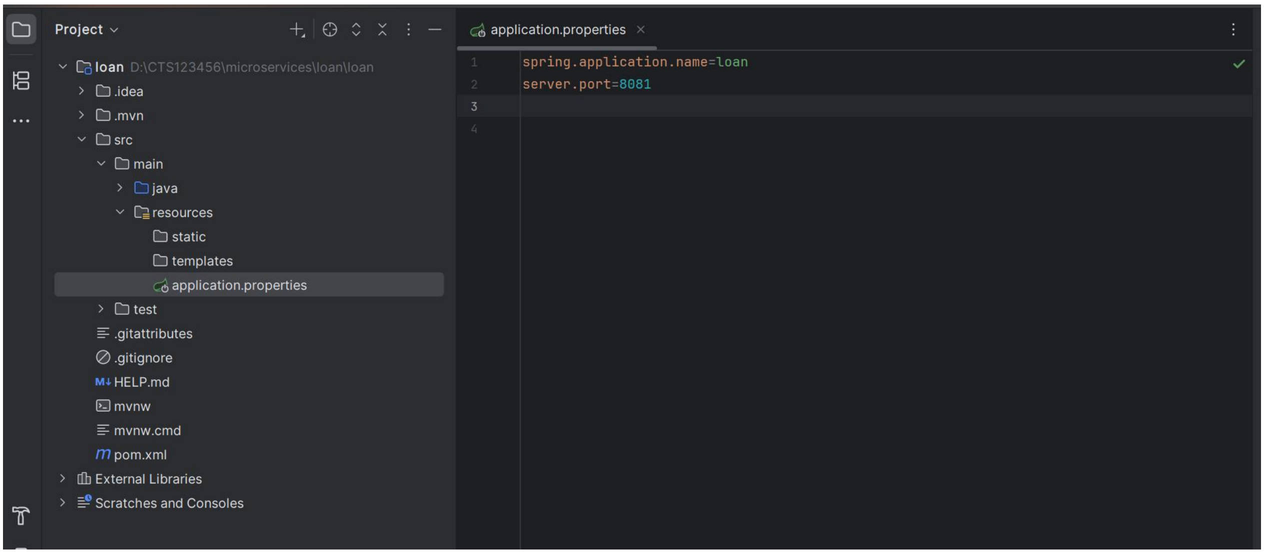
Creating Loan microservice:



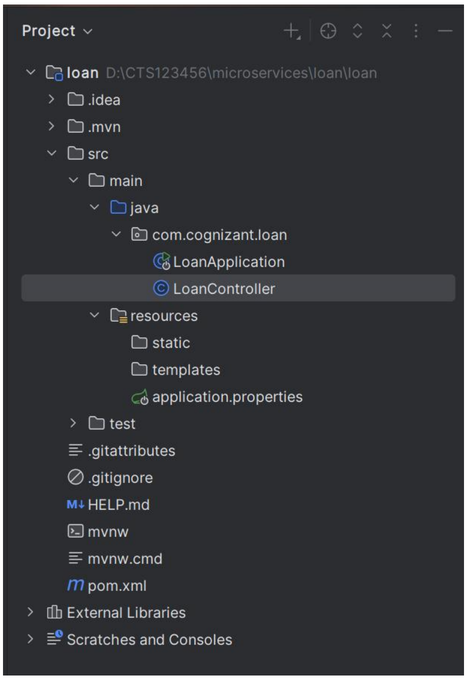
Then move it to the same old folder,



Set custom port for the loans project by adding → server.port=8081



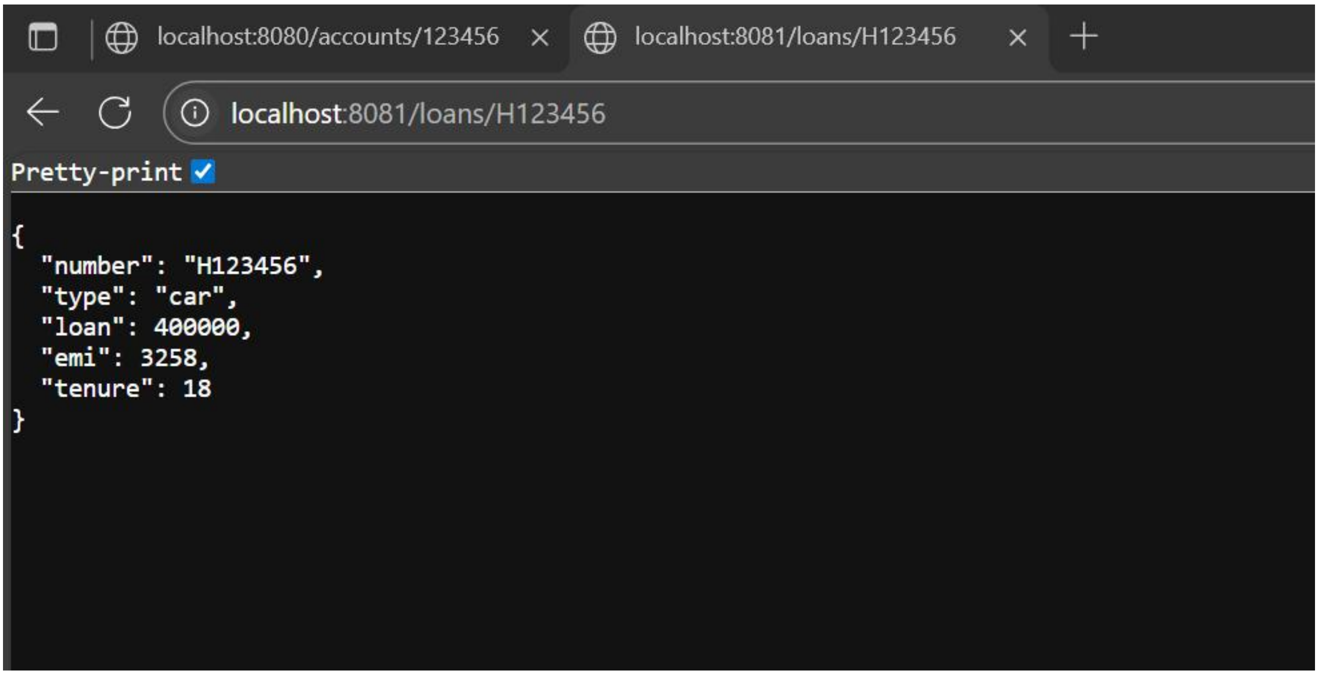
Folder structure:



LoanController.java

```
package com.cognizant.loan;

import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/loans")
public class LoanController {

    @GetMapping("/{number}")
    public Loan getLoan(@PathVariable String number) {
        return new Loan(number, "car", 400000, 3258, 18);
    }

    record Loan(String number, String type, double loan, double emi, int tenure) {}
}
```

Run LoanApplication.java and go to http://localhost:8081/loans/H123456



I now have two microservices,

1) Account on port 8080
2) Loan on port 8081.